

Proposed Visualization & Platform Improvements

Conference Mapping Project – Forward-Looking Recommendations

Overview

This document outlines potential areas of improvement for the conference mapping project, with a focus on information visualization, usability, and sustainability. These recommendations reflect features and design directions that would meaningfully extend the current work and help guide a future team within a short academic development cycle.

1. Making the Sector-Based (Beta) Graph Modes Fully Functional

The current interface includes several sector-based graph modes labeled as beta. These modes would benefit from being made fully functional from data ingestion through network generation and export.

- Event → Sector → Organization (Clean): attendance-only three-layer network.
- Event → Sector → Organization + Connections: adds organization-to-organization ties based on shared participation.
- Organization + Event Only: bipartite network with improved labeling and attributes.
- Organization Connections Only: pure organization-to-organization network.

All modes should enforce consistent schemas, robust validation with clear errors, and reliable exports for both Gephi and Kumu.

2. Deploying and Hosting as an Official Website

The project could be strengthened by transitioning from a prototype deployment to a stable, officially hosted platform intended for repeated academic and research use.

- Stable domain and lightweight branding
- Clear landing page describing purpose and audience
- Versioning and changelog
- Error handling, file-size limits, and browser compatibility
- Privacy note clarifying data handling

3. Improving the Analytics Dashboard

The analytics dashboard represents the strongest opportunity to enhance the project from an information visualization perspective by supporting sense-making before export.

- Network overview metrics (nodes, edges, density, average degree)
- Distribution views (degree and edge-weight distributions)
- Top-entity tables (organizations, events, sectors)
- Interactive filters by year, region, and sector
- Clear explanations of visual encodings
- Data quality indicators (missing values, duplicates)

Summary

These recommendations build on the existing foundation by improving interpretability, interaction, and sustainability. The proposed scope is realistic for a short academic development cycle and offers future teams meaningful opportunities to engage with network visualization and analytical design.